3. Run

3.1 Start the GUI

```
cd jac317/v001
out/gcc/treesync          # or out/clang/treesync
```

Run the **release** build (see 2. Build); `./verify.sh` leaves one in `out/`.

If **Show this page when TreeSync starts** is ticked, the Build page comes first: click **Continue to TreeSync using GCC exe** (or Clang) to go on, or **Build** first.

Options on the command line set up the window and become the saved settings:

```
out/gcc/treesync --source=/home/jay/Photos --target=/run/media/jay/Backup/Photos --mirror
```

Without options, the settings saved last time are used (`~/.config/treesync/settings`, saved on every change).

Run as yourself, not with sudo. The GUI is an ordinary desktop program. When a sync needs root, use **Sync > Run as root** (3.9); only the sync itself runs as root.

3.2 The main window

```
+-------------------------------------------------------------------------------+
| File  View  Sync                    Elapsed 0:12  Total ~0:16  Done 03:34     |
+-----------------------------------+-----------------------------------+
| Disk  [ nvme0n1 ...          v]   | Disk  [ sdb ...                 v]   |
| Mount [ /home/jay            v]   | Mount [ /run/media/jay/Backup   v]   |
| Path  [ /home/jay/Photos ] [...]  | Path  [ .../Backup/Photos ] [...]    |
| [-] [x] Source: /home/jay/Photos  | [-] Target: .../Backup/Photos        |
|     [+] [x] 2024                  |     [+] 2024                         |
|     [+] [-] 2025                  |                                      |
|         [ ] notes.txt             |                                      |
+-----------------------------------+-----------------------------------+
| Pre-scan: source        | Copy                  | Post-scan: target       |
| [==========      ]      | [=====          ]     | [===             ]      |
+-------------------------------------------------------------------------------+
| Copying                              Rate       Size     Time              |
| [ 12/140] 2024/IMG_0001.jpg  [====   ]   85 MiB/s   130 MiB  0:01            |
+-------------------------------------------------------------------------------+
| log: one line per copied file, warnings, the summary                          |
+-------------------------------------------------------------------------------+
| status bar: the command line the current settings stand for                   |
+-------------------------------------------------------------------------------+
```

- **Left: the source**, **right: the target**. Each side has a **Disk** and **Mount** list to jump to a drive, a **Path** box (type a path and press Enter, or **...** to browse), and the tree below it.
- **Trees**: click [+] to open a folder, or use the + and - keys. **View > Reload trees** reads them again, **View > Collapse all** closes them.

- **The progress boxes** and the list of files in progress appear while a run is going. The upper right of the menu bar shows how long it has run, how long the whole run will probably take, and when it should be done.
- **The log** shows what happened; **View > Show log file** (Ctrl+L) opens the full log of the newest run.

Without root, local paths are limited to your home folder; the status bar says so.

Help opens this manual: the whole of it, or one chapter, as a PDF. **F1** opens the chapter for the window you are in.

3.3 Choose what to sync: the checkboxes

Every folder and file in the **source** tree has a checkbox.

- **Check a folder**: everything below it is synced, including files added there later.
- **Uncheck a folder**: nothing below it is synced.
- **Uncheck one file** inside a checked folder: the rest of that folder stays checked.
- A folder with only some of its contents checked shows a **dash** (partly checked).
- The **top box** checks or unchecks the whole tree.

What is checked is saved as `--include-path` settings. Folders above a checked item are created on the target only where something checked goes, and **Mirror never deletes anything outside the checked paths**. Changing the source folder starts again with everything checked. If nothing is checked, Run refuses to start.

3.4 Setup: mode and options

File > Setup... has three tabs.

Source tree / Target tree: the path, and for a remote machine the **Remote** box with **Host**, **Port**, **User** and **Key file** (see 3.10).

Flags:

- **Mode**:
  - **Copy**: new and changed files go from source to target; nothing on the target is removed. Turns on *Skip older*: a newer file on the target is not overwritten.
  - **Mirror**: the target is made to match the source; files only the target has are removed (with safe delete: moved to `<target>.old`).
  - **Merge**: changes go both ways; a file changed on both sides since the last sync is a conflict and is left alone. Nothing is removed.
- **Reset to the defaults** of Copy, Mirror or Merge.
- **Owners**: don't copy / by name (normal) / by number. Setting an owner needs root.
- **Switches**: click any option to see what it does in the help box. The important ones: *Safe delete* (on; keep it on), *Hard links*, *Follow links*, *Skip older / newer / changed*, *Keep extra*, *Wipe target first*, *Low priority*.
- **Numbers**: threads (default 4), retries, chunk size, bandwidth limit, how long to wait for a lost network, time tolerance.

- **Include / exclude**: file and folder names with `*` and `?`, separated by `;`. A blank box filters nothing; exclude wins over include.

3.5 Dry run, then run

1. **Sync > Dry run** (Shift+F9): lists what a run would do (`copy`, `link`, `delete`, `attrs`, `mkdir`) and changes nothing. Read it.
2. **Sync > Run** (F9): asks for confirmation, saying what the mode will do and where removed and replaced files go, then runs.

While it runs:

- **Pre-scan: source** walks the source. **Copy** shows the bytes copied of the total. **Post-scan: target** checks each target folder for files to remove (Mirror).
- Each file is written to `NAME.treesync-part`, read back, checked against the source's SHA-256, given the source's permissions, times, owner and extended attributes, and only then renamed into place.
- A file whose content is the same but whose permissions or time changed gets only its attributes set (an `attrs` step), without copying.

At the end a popup says how it went and how long it took. **Sync > Stop** (Esc) stops after the current file or chunk.

3.6 Stop and resume

A stopped run (Stop, Ctrl+C in a terminal, a crash, a reboot, an unplugged disk) is **resumed by running it again with the same settings**. Every step was recorded in the database, so it continues where it was. A big file continues at its last good chunk: on resume, every chunk already copied is read back and checked first, and from the first damaged one the file is copied again.

To start over instead, use `--fresh` on the command line.

3.7 Checking and attributes

- **Sync > Scan only (SHA-256 check)** (F7): changes nothing. Lists both trees, reads and hashes **every file on both sides** afresh, and compares permissions, times, owners and extended attributes. Each file's log line ends in `same`, `DIFFERS`, `missing` (not on the target) or `failed`. Use it after a big copy to prove the target is right.
- **Sync > Sync attributes** (F8): sets only the attributes that differ (mode, time, owner, extended attributes) on the target, for files of the same size and for folders. No content is copied.
- **Sync > Clear .old folders...**: deletes the safe-delete folders (`<target>.old`) for good, after a confirmation that says how many files and bytes. Folders made by a root run need **Run as root**.

3.8 History, logs and AI export

- **View > History...** (Ctrl+H): every run, newest first, with its counts (folders, copied, deleted, attributes set, failed, conflicts, bytes). Failed runs are red. Below:
  - **Actions**: what the selected run did, failures first.
  - **SHA-256 checks**: every checked file under a line naming its folder, with the source and target SHA-256 side by side, whether they match, and how the attributes compare. Mismatches first, in red; attribute differences in orange.
  - **Show log**, **Export logs to AI.md**, **Export SQL to AI.md**.
- **Logs**: each run writes `logs/YYYY-MM-DD_HHMMSS.log` next to the database. It starts with a header; every line has its start, stop and duration to the microsecond.
- **View > Export newest run for AI**: writes the run as Markdown files into `logs/ai/` (a guide, problems, slowest copies, throughput, the logs in full; or the database contents), ready to give to an AI to analyse.

3.9 Running as root

A sync often meets files your user cannot read or replace (other users' files, system files). Without root, local trees must be inside your home folder.

- **In the GUI**: tick **Sync > Run as root (pkexec)**. Each run then starts `treesync-cli` through `pkexec`, which asks for your password; the GUI itself stays an ordinary program.
- **In a terminal**: `sudo out/gcc/treesync-cli ...`.

A root run keeps its own database and logs under `/root/.local/share/treesync/`, separate from yours. To resume a root run, run it as root again.

3.10 Remote trees (SFTP)

Either side can be on another machine over SSH. One-time setup:

```
ssh-keygen -t ed25519            # if you have no key yet
ssh-copy-id jay@server           # passwordless login
ssh jay@server                   # once, to check and accept the host key
```

TreeSync logs in with keys only, never passwords. In the GUI: **File > Setup > Target tree**, tick **Remote**, and fill in Host, Port, User (and Key file if not the usual one). On the command line: `--target-host=server --target-user=jay`.

If the network drops, the run waits (default up to 30 minutes, *Wait for the network*), shows **NETWORK UNAVAILABLE**, and carries on when the machine is back.

3.11 The command line

`treesync-cli` takes the same options the GUI saves. **It does a dry run unless you give `--no-dry-run`.**

```
# What would a mirror do?
out/gcc/treesync-cli --source=/home/jay/Photos --target=/run/media/jay/Backup/Photos --mirror

# Do it
out/gcc/treesync-cli --source=/home/jay/Photos --target=/run/media/jay/Backup/Photos --mirror
--no-dry-run

# Copy only new and changed files, 8 threads, at low priority
out/gcc/treesync-cli --source=/data --target=/backup/data --copy --threads=8 --low-priority -
-no-dry-run

# Only one folder and one file of the source
out/gcc/treesync-cli --source=/data --target=/backup/data --include-path=projects/2026 --
include-path=notes.txt --no-dry-run

# To a remote machine
out/gcc/treesync-cli --source=/data --target-host=server --target-user=jay --
target=/backup/data --no-dry-run

# Prove a copy: hash both sides, change nothing
out/gcc/treesync-cli --source=/data --target=/backup/data --scan-only

# Set only differing permissions and times
out/gcc/treesync-cli --source=/data --target=/backup/data --sync-attrs --no-dry-run

# Delete <target>.old for good (without --no-dry-run it only counts)
out/gcc/treesync-cli --source=/data --target=/backup/data --clear-old --no-dry-run
```

`treesync-cli --help` lists every option. In a terminal it shows one line per copied file, `[ 6/90] name 100% | 85.3 MiB/s | 130.9 MiB | 00m01s`, and Ctrl+C stops it (press twice to quit at once).

Exit codes: 0 in sync (or, for Scan only, the trees are the same); 1 failures, conflicts or differences; 2 could not run (bad options, a path refused); 3 stopped.

Database views for looking at results directly:

```
sqlite3 ~/.local/share/treesync/treesync.db
```

```
SELECT run, started, state, copied, failed FROM run_summary ORDER BY run DESC LIMIT 10;
SELECT * FROM failures WHERE run = 42;
SELECT path, file_name, problem FROM hash_mismatches WHERE run = 42;
SELECT path, file_name, attr_diff, result FROM attr_mismatches WHERE run = 42;
```